「Def.

In a Ring R, an ideal $M \subsetneq R$ is called a Maximal ideal if:

there is no ideal N such that $M \subsetneq N \subsetneq R$.

Equivalently, if an ideal $N \subseteq R$ satisfies $M \subseteq N$, then either $N = M$ or $N = R$.

「Def.

In a Commutative Ring R, an ideal $P \subsetneq R$ is called a Prime ideal if:

for any $a, b \in R$, if $ab \in P$, then either $a \in P$ or $b \in P$.

Equivalently, if $(ab) \subseteq P$, then $(a) \subseteq P$ or $(b) \subseteq P$.

「Prop.

Let R be a Commutative Ring, and $M, P \subsetneq$ be proper ideals.

M is a Maximal ideal if and only if R/M is a field

P is a Prime ideal if and only if R/P is an integral domain.

Proof. To show the first assertion, notice that:

An ideal $I \subseteq R$ contains a unit if and only if $I = R$, because:

given $r \in R$, $r = r \cdot 1 = r \cdot v^{-1} \cdot v \in I$. And, the converse is trivial since $1 \in R = I$.

And, every non-zero element of field is unit, so

F is field if and only if if $I \subseteq F$ is an ideal, then either $I = \{0\}$ or $I = F$.

Now, $M \subsetneq R$ is maximal ideal if and only if there is no ideal N s.t. $M \subsetneq N \subsetneq R$.

4th iso. theorem ↑ if and only if there is no non-trivial proper ideal of R/M

if and only if R/M is a field.

And, $P \subsetneq R$ is prime ideal if and only if $ab \in P \Rightarrow a \in P$ or $b \in P$

if and only if $ab + P = P \Rightarrow a + P = P$ or $b + P = P$.

if and only if $\bar{a} \cdot \bar{b} = 0 \Rightarrow \bar{a} = 0$ or $\bar{b} = 0$

if and only if $\bar{R} = R/P$ has no zero divisor

So proved. ☑